

Website/App User Funnel Drop-off Analysis

Data Analyst Intern Assessment — Submission Report

1. Objective

Given event-level signup/checkout data (user_id, step, timestamp), calculate the conversion rate at each stage of the funnel, identify the stage with the biggest drop-off, and present the findings in a form a non-technical stakeholder could act on.

2. Dataset & Methodology

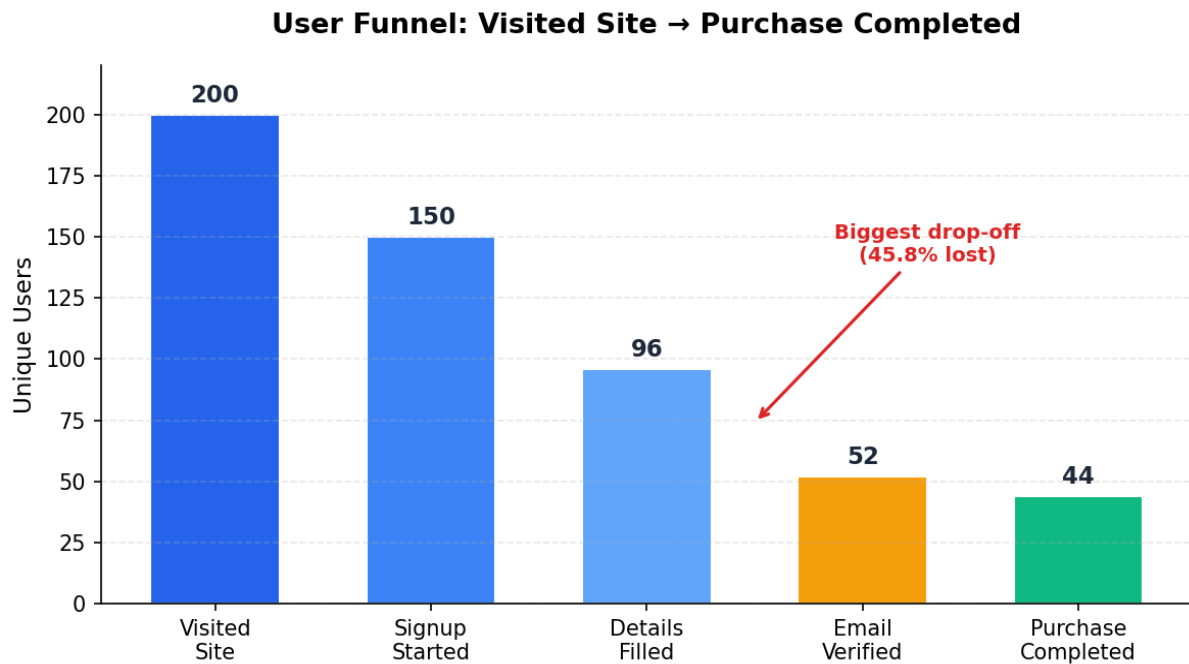
The dataset contains 542 event rows covering 200 unique users across five ordered funnel stages: visited_site → signup_started → details_filled → email_verified → purchase_completed.

Steps taken:

- Parsed timestamps and checked for duplicate (user_id, step) events — none were found, so no de-duplication was needed beyond keeping the earliest timestamp per user/step as a safeguard.
- Counted the number of **unique users** who reached each stage (a user reaching a stage counts once, regardless of repeat events).
- Calculated **stage-to-stage conversion rate** = (users at current stage) ÷ (users at previous stage) × 100.
- Calculated **overall conversion** = (users at current stage) ÷ (users at visited_site) × 100, to show cumulative reach.
- Compared drop-off in both absolute users lost and percentage lost to identify the true bottleneck.

3. Funnel Breakdown

Stage	Unique Users	Conv. from Prev. Stage	Users Lost	Drop-off %	% of Total
Visited Site	200	—	—	—	100.0%
Signup Started	150	75.0%	50	25.0%	75.0%
Details Filled	96	64.0%	54	36.0%	48.0%
Email Verified	52	54.2%	44	45.8%	26.0%
Purchase Completed	44	84.6%	8	15.4%	22.0%



4. Biggest Drop-off

■ Details Filled → Email Verified

This is the single biggest leak in the funnel. Only **54.2%** of users who filled in their details went on to verify their email — a loss of **44 users (45.8%)**, the worst conversion rate of any step. In absolute terms, the Signup → Details Filled step loses slightly more users (54 vs. 44), but Details Filled → Email Verified has the worst *rate* of loss, meaning proportionally it is where the funnel is weakest and the highest-leverage point to fix first.

5. Data Quality Observations (Bonus)

No duplicate events were found (each user/step combination appears once). However, 146 of 200 users have a non-contiguous stage history — e.g. a user with only visited_site, signup_started, and email_verified logged, but no details_filled event. This is treated as an implicit skip/logging gap rather than an error: the funnel counts here use each user's **furthest reached stage**, which is the standard approach for weekly funnel reporting and matches how the conversion rates above were calculated. If this were a real production pipeline, I would flag the missing details_filled events to engineering, since it could indicate an analytics tracking gap rather than an actual product drop-off.

6. Time-to-Convert (Bonus)

Stage Transition	Avg. Time to Convert
Visited Site → Signup Started	15.3 minutes
Signup Started → Details Filled	13.6 minutes

Note: average time from Details Filled → Email Verified and Email Verified → Purchase Completed could not be reliably computed for most users because of the logging gaps described above (many users are missing the intermediate details_filled timestamp).

7. Recommendation

Prioritize investigating the **Details Filled** → **Email Verified** step. Likely causes to test: the verification email landing in spam, an unclear or delayed email trigger, or friction in the verification UI itself. A/B testing a resend prompt or reducing the wait time between form submission and email delivery would be the first experiment I'd run.

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Tools used: Python (pandas) for aggregation, matplotlib for visualization, reportlab for this report.